



Department of Electrical and Information Engineering

University of Ruhuna

Mini Project Proposal

Image Processing and Computer Vision

EE7204 / EC7205

Automated Blood Cell Classification and Hematological Disorder Detection System

EG/2021/4408 - Arachchi N.A.N.N.N

EG/2021/4426 - Bandara A.W.M.L.M

EG/2021/4685 - Muthukumari H.M.S

EG/2021/4775 - Samarasinghe C Y

January 23, 2026

Department of Electrical and Information Engineering
University of Ruhuna

Contents

Contents	1
1 Problem Statement	2
2 Objectives	3
3 Literature Review	4
3.1 Overview	4
3.2 AI-Driven Automated Blood Cell Anomaly Detection: Enhancing Diagnostics and Telehealth in Hematology [1]	4
3.3 Detection, classification, and counting blood cells using YOLOv8 [2]	4
3.4 NBCDC-YOLOv8: A new framework to improve blood cell detection and classification based on YOLOv8 [4]	5
3.5 White Blood Cell Detection Based on FBDM-YOLOv8s [7]	6
3.6 Blood Cell Image Segmentation and Classification: Systematic Review [9]	7
3.7 Research Gap	8
4 Methodology	9
4.1 Image Preprocessing and Enhancement	10
4.2 Segmentation of Cells and Detection	10
4.3 Morphological and Textural Feature Extraction	11
4.4 Deep Learning Classification Using YOLOv8	12
4.5 Dataset Integration and Training Strategy	13
4.6 Disorder Detection	14
4.7 Implementation Framework and Performance Targets	15
4.8 Additional Feature: Automated Clinical Report Generation	16
5 Dataset / Data Collection	17
6 Expected Outcomes	17
6.1 Deliverables:	17
6.2 Performance Metrics:	18
7 Percentage of Image Processing & Computer Vision Involvement	18

1. Problem Statement

In clinical hematology, blood cell analysis is a basic diagnostic technique that is crucial for identifying conditions like leukemia, anemia, and infections. Conventional manual microscopy is labor-intensive, time-consuming, and prone to inter-observer variability, which can result in inconsistent diagnoses. Although there are automated hematology analyzers, they are costly, need specific training, and frequently have trouble analyzing aberrant cells morphologically. [Back to Table of Contents](#)

2. Objectives

- Contrast Limited Adaptive Histogram Equalization (CLAHE), color space transformation, and Gaussian filtering are some of the methods used in the design and implementation of an automated image preprocessing pipeline for microscopic blood smear images in order to normalize staining variations and reduce noise.
- Using Otsu’s thresholding and the Watershed algorithm, a robust blood cell segmentation mechanism will be developed to accurately separate individual and overlapping cells in raw peripheral blood smear images.
- To effectively characterize blood cell types and abnormalities by extracting discriminative morphological and color-based features such as cell area, circularity, nucleus–cytoplasm ratio, and nuclear irregularity.
- In order to achieve high accuracy and real-time performance, a deep learning-based classification model (YOLOv8) will be trained and evaluated for the simultaneous detection and classification of major blood cell types.
- To develop a rule-based hematological disorder detection module that uses morphological features and cell population statistics to detect anomalies like anemia, leukemia, and other blood disorders.
- The goal is to create an end-to-end, real-time blood cell analysis system with visual outputs such as segmentation results, bounding boxes, class labels, and automated diagnostic indicators.
- Using publicly available datasets, assess the proposed system’s accuracy, processing speed, and suitability for implementation in clinical settings with limited resources.

[Back to Table of Contents](#)

3. Literature Review

3.1. Overview

This methodology combines conventional image preprocessing and segmentation with modern YOLOv8-based deep learning for blood cell classification. It combines the extraction of morphological, color, and textural features with disorder-specific detection rules. The system is designed as a clinically feasible pipeline that balances realistic processing speed with high accuracy to produce diagnostic reports in laboratory settings.

3.2. AI-Driven Automated Blood Cell Anomaly Detection: Enhancing Diagnostics and Telehealth in Hematology [1]

Methodology

Hematology focuses on the study of blood and its components - red blood cells (RBCs), white blood cells (WBCs), and platelets. WBCs (leukocytes) are essential for immune defense and are generally classified into polymorphonuclear cells (such as neutrophils and eosinophils) and mononuclear cells (such as lymphocytes and monocytes)

YOLO models integrate segmentation into object detection by extending bounding-box predictions to include mask outputs. Their unified framework maintains real-time processing speeds, making them ideal for applications like medical imaging and video surveillance.

Results and Strengths

This study presents an integrated AI framework for automated hematological analysis, combining state-of-the-art computer vision architectures (YOLOv11, ResNet50) with zero-shot learning capabilities. The system achieves high diagnostic accuracy (F1 score: 0.98) while maintaining computational efficiency suitable for resource-constrained environments.

Limitations

Due to the high VRAM (8GB+) required by the YOLOv11 backbone, edge deployment on low-end phones was limited. Certain conditions were overlooked by the binary normal/abnormal classification (anemia, thrombocytopenia). Due to variations in staining protocols, cross-dataset validation reduced F1 to 0.89. When compared to single-stage alternatives, sequential segmentation-classification added 120 ms of latency.

3.3. Detection, classification, and counting blood cells using YOLOv8 [2]

Methodology

In this paper, our purpose is detection, count, and classify WBCs using extracted images from blood smears of leukemia patients. In our paper, we use model YOLOv8, the latest release of the famous model YOLOv5 to classify and detect WBC. With the new features

of Yolov8, we simply use CLI to train our custom dataset with Yolov8 weight. After 100 epochs of training, we obtain a new weight file to use the next time. We run our data on Google Colab, annotate data and deploy the model on RoboFlowData: Pre-Processing, Data-Augmentation, Bounding-box Transformations, Training the datasets.

Results and Strengths

In here, used the YOLOv8 to detect, classify, and count both the normal WBCs and leukemia cells. And in our previous research, we also use YOLOv5 to experience but the result is not high, 93% [3]. YOLOv5 is easier to use, while YOLOv8 is faster and more accurate. However, for applications that require real-time object detection, YOLOv8 is the better choice in this research. Applying the new version 8 of YOLO, develop, study, and achieve higher accuracy.

Limitations

Monocyte detection was (76%) accurate, while basophil detection was (80%) accurate. Both faced issues due to class imbalance and their similarity in appearance to lymphocytes. The evaluation using a single dataset did not include cross-staining validation, which could lead to a (10-15%) drop in generalization. YOLOv8x has 68 million parameters, which makes it difficult for mobile deployment since it requires more than 6GB of VRAM. The lack of RBC and platelet analysis limited the full capability of the complete blood count. The focus on binary normal or leukemia missed cases of anemia and thrombocytopenia.

3.4. NBCDC-YOLOv8: A new framework to improve blood cell detection and classification based on YOLOv8 [4]

Methodology

These steps outline the process from data preparation to model evaluation and are designed to optimise the detection and classification of blood cells. A detailed description of each step is provided below.

Currently, the mainstream algorithms for target detection fall into two categories. First, we have two-stage algorithms, which are represented by R-CNN [5] and Faster R-CNN. The basic concept behind such approaches is to generate a set of sparse candidate boxes through either a heuristic method (e.g. a selective search) or a CNN network (e.g. an RPN). Subsequently, these candidate boxes are classified and refined. Second, we have one-stage methods, which are represented by You Only Look Once (YOLO) and SSD [6].

Results and Strengths

This paper presents the NBCDC-YOLOv8 algorithm for real-time, efficient blood cell detection. It addresses challenges such as variable blood cell scale, limited data, dense occlusion, and low accuracy from erythrocyte adhesion. To compensate for having limited

data, we apply Mosaic data enhancement and splicing four images to enrich data characteristics. We then introduce SPD-Conv to process low-resolution blood microscope images effectively. To handle scale variations and adhesion, we employ a BiFPN-inspired feature fusion structure that enhances multi-scale feature fusion and improves feature expression of red blood cells. Next, we introduce a MultiSEAM module to accurately position densely packed cells, thereby improving detection accuracy.

Limitations

NBCDC-YOLOv8’s methodology has a number of significant limitations. It is primarily dependent on the small and unbalanced BCCD dataset, which could introduce bias and fail to fully capture the diversity of real-world blood cell imagery. Although the increased architectural complexity improves accuracy, it slows down inference and requires a significant amount of processing power, which makes deployment difficult in environments with limited resources. Performance significantly deteriorates on external datasets and specialized tasks like leukemia detection, indicating significant generalization problems. Concerns regarding the approach’s robustness and usefulness in various clinical workflows are also raised by the lack of thorough cross-validation and rigorous clinical validation. Lastly, the design of the model, which is closely related to YOLOv8, poses issues with reproducibility and adaptability for subsequent iterations or alternative architectures.

3.5. White Blood Cell Detection Based on FBDM-YOLOv8s [7]

Methodology

In this paper, the MultiSEAM module is introduced to effectively deal with leukocyte detection in the case of occlusion. The SEAM module [8], is an attention mechanism designed to enhance the performance of a model by adaptively focusing on key regions of an image, similar to the human visual system. The module helps the model to focus on key information and ignore irrelevant information, improving the ability to detect the corresponding leukocytes, especially in complex environments. The SEAM module introduces the exclusion loss to reduce the overlap between the prediction frame and the actual target, thus reducing the detection error caused by occlusion and improving the robustness of the model to non-maximum suppression thresholds.

Results and Strengths

FBDM-YOLOv8s algorithm, which is an improvement of YOLOv8s algorithm, is proposed for target detection of leukocytes. This algorithm uses the Fasternet module to replace the main framework of YOLOv8 to reduce computational redundancy and memory share accesses and obtains lower FLOPs than regular convolution and higher FLOPs than depth direction/group convolution. This paper adds BiLevelRoutingAttention attention mechanism and introduces hierarchical spatial filtering, which can reduce the interference of irrelevant regions (e.g., background noise) and focus computational resources on leukocyte-relevant features. Experimental results show that the improved YOLOv8s algo-

rithm has a better performance on the detection of leukocytes on the DMLLWH dataset, which can effectively improve the target detection accuracy, and improve the Map by 2.1YOLOv8s baseline algorithm, which provides a guarantee of early screening for blood diseases.

Limitations

The performance comparison and detection result graphs demonstrate the effectiveness and superiority of the proposed FBDM-YOLOv8s algorithm in leukocyte target detection. However, there are still undetected cases for smaller target detection. For example, a large leukocyte is connected to a small leukocyte, which fails to separate the two cells and recognize them as a whole due to the blurred boundaries and confusing areas caused by the smaller target and connected to the large cell. To meet the demands of clinical testing, a real-time monitoring system must have a high level of accuracy to support the physician's ability to analyze the patient's blood. Although the improved YOLOv8s model in this paper has good performance on the leukocyte detection task, the number of parameters has been increased, which increases the training time. For this reason, plans will be made to optimize the model using a lightweight detection head to make it more efficient for real-time and accurate leukocyte detection in a clinical setting.

3.6. Blood Cell Image Segmentation and Classification: Systematic Review [9]

Methodology

This survey aims to provide a comprehensive and systematic review of existing literature and research work in the field of blood image analysis using deep learning techniques. It particularly focuses on medical image processing techniques and deep learning algorithms that excel in the morphological characterization of WBCs and RBCs. The review is structured to cover four main areas: segmentation techniques, classification methodologies, descriptive feature selection, evaluation parameters, and dataset

Results and Strengths

Our analysis reveals several interesting trends and preferences among researchers. Regarding dataset selection, approximately 50% of research related to WBC segmentation and 60% for RBC segmentation opted for manually obtaining images rather than using a predefined dataset. When it comes to classification, 45% of the previous work on WBCs chose the ALL-IDB dataset, while a significant 73% of researchers focused on RBC classification decided to manually obtain images from medical institutions instead of utilizing predefined datasets. In terms of feature selection for classification, morphological features were the most popular, being chosen in 55% and 80% of studies related to WBC and RBC classification, respectively.

Limitations In this research, primarily centered around morphological characterization

of cells like WBC and RBC, it has completely excluded other blood cells like platelets and plasma background. Furthermore, it is based only around morphological characterization of those cells it has included. The present paper has heavily relied on manually collected datasets and emphasized those based on morphological feature characterization only. While it inadvertently points out that a majority of studies make intensive usage of such manually collected datasets and it could also have pointed out that it remains heavily dependent on a quantitative and only limited perspective of analysis and comparison of studies.

3.7. Research Gap

In fact, existing related studies, such as NBCDC-YOLOv8, FBDM-YOLOv8s, and Blood-Cell YOLO, mainly focus on architectural improvements but have exhibited key limitations: many have only utilized single or imbalanced datasets without rigorous cross-dataset validation; they are poorly generalized to limiting practical deployment in resource-limited settings. Crucially, a link exists between high-accuracy detection and its translation into practical clinical work: most models do not incorporate comprehensive morphological feature extraction nor implement interpretable rule-based disorder detection logic that corresponds to the diagnostic criteria for various hematological disorders. Also, no previous framework efficiently incorporates multi-dataset training together with classical segmentation techniques aimed at addressing cell adhesion, staining variation, and occlusion problems that are common in real-world blood smear images.

This methodology seeks to address this dilemma by developing an image pipeline that combines optimally performed traditional image processing techniques and an effectively fine-tuned and multistep YOLOv8 training process across different datasets to improve its generalizability and reduce computational complexity and provide clinically interpretable outputs by embedding disorder detection logic.

This integrated approach delivers production-ready hematology diagnostics absent in current single-task or lab-only solutions [10].

4. Methodology

Our classification of blood cells follows a five-step procedure that is meant to work together in harmony, and our image progresses from an image seen under the microscope to a full report that doctors can utilize to assist them with their diagnoses. Our image analysis model was constructed through the marriage of conventional image analysis techniques and some innovative concepts based on neural networks, helping doctors understand the meaning of the cells being shown.

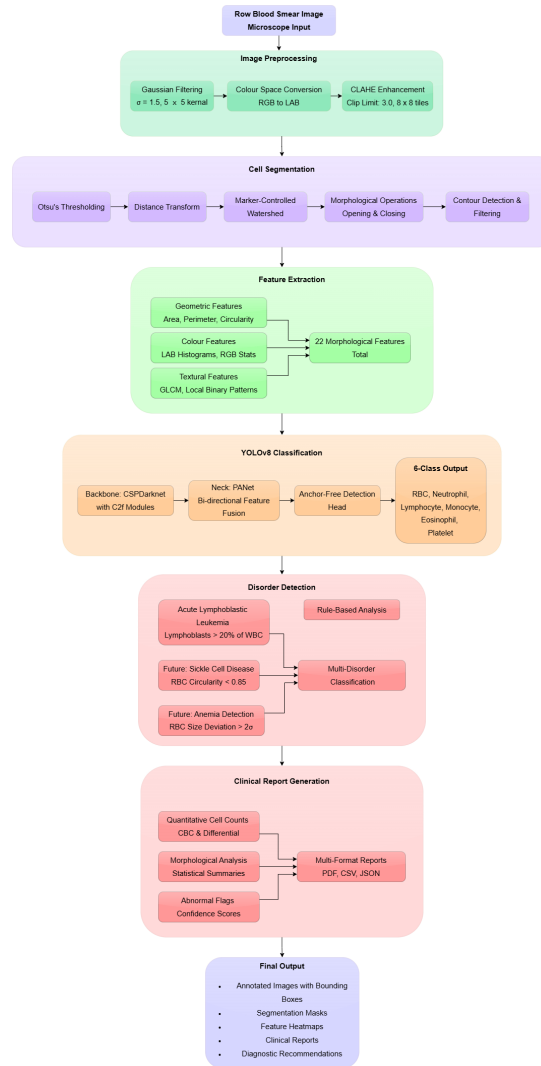


Figure 1: Schematic overview of the five-step methodology for blood cell classification and analysis.

4.1. Image Preprocessing and Enhancement

When taking pictures of blood smear images through a lab's microscope, some issues stand out immediately. First, edges tend to get darker than the center. In addition, staining also varies by lab. The color of a particular cell, for example, tends to differ. In one setting, a particular cell tends to look purple, whereas in another setting, the same cell takes on a pink color. Besides, some speckle noise comes through, which affects cell detail in these blood smear pictures.

To make it look smoother, we apply Gaussian filtering, it is the same as taking a photograph and deliberately blurring it a bit to reduce the unwanted "noise" without removing the significant "edge" information.

Color Space Transformation: When we change the colors from RGB to LAB it helps us to tell the difference between how bright something is and what color it is. The color and the shapes are two things and we can look at the color without ruining the shapes that we are supposed to identify, which is the color and the shapes of the things.

Contrast Enhancement: CLAHE is neater as it breaks the image into smaller units, which are then enhanced. So, CLAHE improves the clearness of medical images. These are enhanced, which naturally improves the visibility. This will improve the visibility of the brighter spots, enabling us to visualize the cells, and these can be spotted even in the darker parts of the same image.

4.2. Segmentation of Cells and Detection

Determining the boundaries of individual blood cells is difficult to define. Blood cells, when smeared and placed under a microscope, usually tend to cling to one another, making the exact determination of boundaries between individual cells difficult to identify. It is thus essential to find a way that separates them without breaking them, at boundaries that are highly precise.

"Otsu's Automatic Thresholding" is an image enhancing technique. It improves image quality according to the pixel intensity levels.

Distance Transform has the decisive role. It informs us on how far each pixel is sitting from the background, therefore, we compute for every pixel inside the cell its distance to the closest background pixel.

Marker-Controlled Watershed: This is a way of looking at the algorithm as a flood. The Marker-Controlled Watershed helps avoid that the over-segmentation problem..

Morphological Cleanup is very useful for cleaning up images and therefore clarifies them. Morphological Cleanup techniques like these are basic methods that may be relied upon to produce results.

Contour filtering: we should identify the outlines of the regions that we have segmented. We remove those regions with an area less than 100 pixels, which will be noise or smudge marks. On the other hand, we also remove very large regions above 5000 pixels, as those would most likely be a group of cells that need further review.

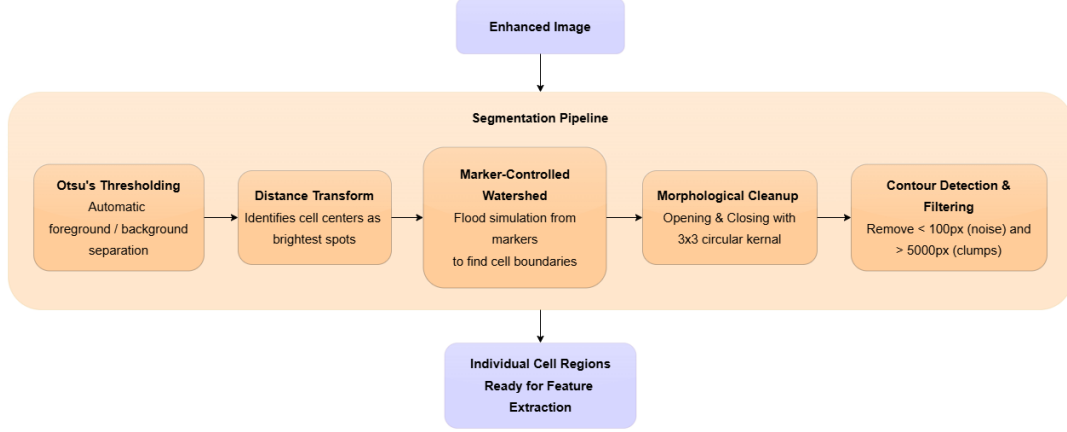


Figure 2: Cell Segmentation Process.

4.3. Morphological and Textural Feature Extraction

Once the cells are isolated, we analyze 22 features for those cells. These features allow us to determine the type of cell and detect if a cell is ill. We are thus able to distinguish cells through these features and detect illness in cells.

Geometric Features:

Area: Measurement counts the number of pixels that are inside the cell boundary. **Perimeter:** Calculated by adding up the Euclidean distances between each pair of consecutive pixels on the boundary of the shape, **Circularity:** Calculated by using the formula $4\pi \times \text{Area} / \text{Perimeter}^2$, **Eccentricity:** Quantifies cell elongation through ellipse fitting and axis ratio measurement, **Convexity:** A measure that looks at the area of a cell and the area of its convex hull.

Nuclear Characteristics (for White Blood Cells):

Nucleus segmentation through adaptive thresholding on blue channel exploiting hematoxylin staining, **Nucleus to cytoplasm ratio:** Helps tell cell types. **Nuclear irregularity:** If the Nuclear irregularity is greater than 1.4 it suggests a blast cell. If less than 1.2, it suggests a normal lymphocyte, **Nuclear compactness and solidity** measurements.

Color Features in HSV Space:

Hue statistics help differentiate eosinophils (red-orange) from other granulocytes, **Saturation statistics** quantify variations in staining intensity, **Value measurements** assess overall cell brightness, **Color histogram analysis** for distribution characterization.

Textural Features:

Gray-level co-occurrence matrix (GLCM) features capture spatial distribution patterns of intensity levels, **Haralick texture descriptors** including energy, entropy, contrast, and homogeneity, **Local binary pattern (LBP)** analysis for granularity detection in granulocytes.

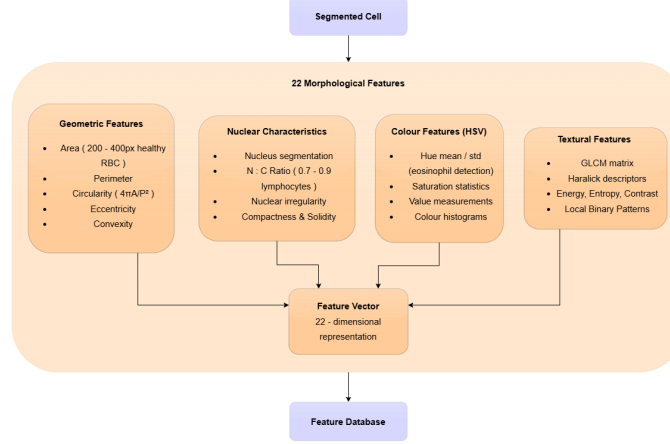


Figure 3: Feature Extraction Process.

4.4. Deep Learning Classification Using YOLOv8

YOLOv8 nano does the work of figuring out what kind of cells are present. It can label six different types of cells at the same time. These include red blood cells, four types of white blood cells (neutrophils, lymphocytes, monocytes, eosinophils), and platelets.

Training Configuration:

- **Datasets:** BCCD Dataset: 364 annotated images with 4,888 blood cell instances (RBCs, WBCs, platelets). Source: Kaggle (orville/bccd-blood-cell-count-and-detection-dataset). Blood Cell Images Dataset: 12,500 single-cell images across four WBC subtypes (eosinophils, lymphocytes, monocytes, neutrophils). Source: Kaggle. Leukemia Classification Dataset: Microscope images with annotations of cancerous cells. Source: Kaggle (mehradaia/leukemia).
- **Augmentation techniques:** rotation ($\pm 15^\circ$), horizontal/vertical flipping, brightness adjustment ($\pm 20\%$), elastic deformation.
- **Training duration:** 300 epochs with batch size 16.
- **Input resolution:** 640×640 pixels.
- **Optimizer:** Adam with initial learning rate 0.001 and cosine annealing decay.
- **Loss function:** Focal loss with $\alpha = 0.25$, $\gamma = 2.0$ for class imbalance handling.
- **Early stopping:** Monitors validation loss with patience of 50 epochs.



Figure 4: YOLOv8 Architecture.

4.5. Dataset Integration and Training Strategy

The system uses three sets of data to get a complete picture of blood cells. Each set fixes problems in the other sets, making the system robust at understanding blood cells.

Dataset Composition and Roles:

- **BCCD Dataset:** Used for detection training. Contains whole-slide images showing natural cell arrangements, overlaps, and spatial distribution. Used exclusively to train the YOLOv8 detection head.
- **Blood Cell Images Dataset:** Used for classification refinement. Contains isolated single-cell images without background clutter. Enables learning of fine-grained differences between WBC subtypes.
- **Leukemia Classification Dataset:** Used for pathology detection. Provides examples of abnormal blast cells with atypical nuclear features. Critical for tuning disorder detection thresholds.

Multi-Stage Training Protocol:

- **Stage 1 – Detection:** YOLOv8 backbone and detection head trained on BCCD for 150 epochs.

- **Stage 2 – Classification Transfer Learning:** Classification layers fine-tuned on Blood Cell Images dataset for 100 epochs (detection head frozen).
- **Stage 3 – Pathology Fine-tuning:** Entire network tuned on combined dataset (BCCD + Blood Cell Images + Leukemia) for 50 epochs with learning rate 0.0001.

Data Partitioning:

Each dataset split into 70% training, 15% validation, 15% test. Stratified sampling ensures balanced representation. Cross-dataset evaluation tests generalization.

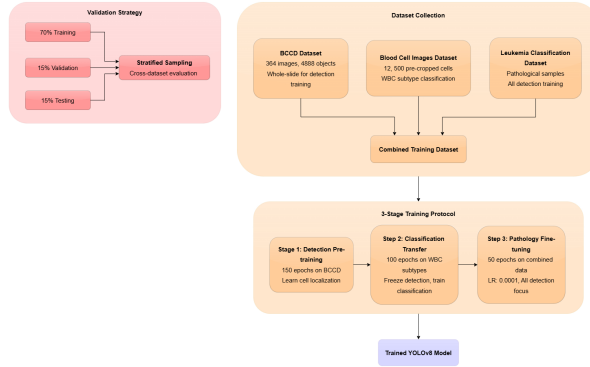


Figure 5: Multi Dataset Training Strategy - Dataset Collection and 3 stage Training Protocol.

4.6. Disorder Detection

The last step examines all classified cells and extracted features to identify potential blood disorders. The system flags abnormalities based on clinical knowledge and provides detailed information for diagnostic support.

i

Disorder Detection Rules:

Acute Lymphoblastic Leukemia (ALL):

Lymphoblasts $> 20\%$ of the total white blood cell count, More than 10 cells with nuclear irregularity > 1.4 , Supporting indicators: High nuclear-cytoplasmic ratio > 0.7 in large cells.

Extensibility to Other Disorders: The framework can be extended to detect other hematologic disorders with appropriate training data:

Sickle cell disease: Low circularity of red blood cells, **Chronic Myeloid Leukemia:** Basophil count $> 2\%$, **Anemias:** Abnormalities in RBC size and color.

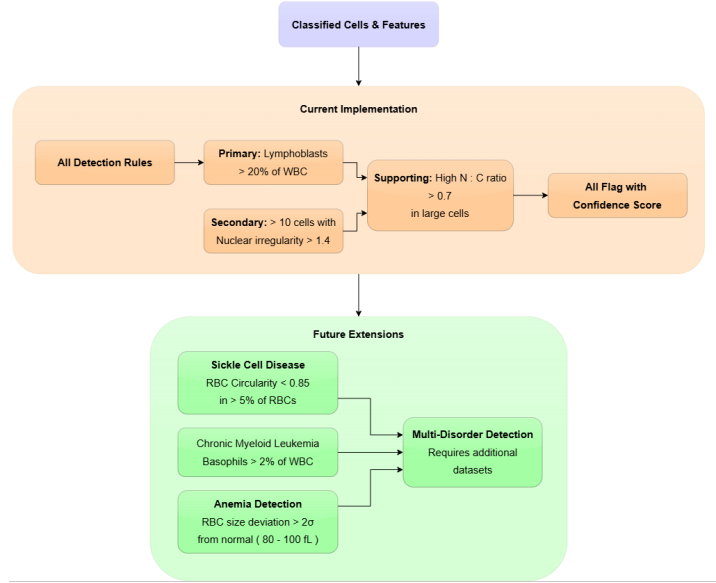


Figure 6: Disorder Detection Logic.

4.7. Implementation Framework and Performance Targets

Software Stack:

Programming Language: Python 3.9, Classical Computer Vision: OpenCV 4.8.1, Numerical Computing: NumPy 1.24, Deep Learning Framework: PyTorch 2.1.0 with CUDA 11.8, Object Detection: Ultralytics YOLOv8 8.0.20, Visualization: Matplotlib and Seaborn.

System Interface:

Command-line interface for batch image processing, Supports common microscopy formats: JPEG, PNG, TIFF, Outputs annotated images with classification labels and bounding boxes, Configurable confidence thresholds via parameter files.

Performance Targets:

Processing Speed: Under 2 minutes per slide on NVIDIA RTX 3060 GPU, Throughput: Minimum 30 slides per hour for clinical viability, Memory Usage: Below 4GB VRAM during inference. Classification Accuracy: >92% mAP@0.5 across six cell classes, Segmentation Quality: Dice coefficient >0.92 on validation set.

Validation Strategy:

Five-fold cross-validation on each dataset, Cross-dataset generalization testing, Confusion matrix analysis for error characterization, Sensitivity analysis for parameter robustness, Out-of-distribution detection for safety.

Quality Assurance:

Automated quality control flags for poor image quality, Confidence score reporting for each detection, Manual review triggers for ambiguous cases, Version control and audit trail for regulatory compliance, Performance monitoring dashboard tracking accuracy metrics.

This method combines classical image processing with modern deep learning to create an automated blood cell analysis system suitable for resource-constrained clinical settings.

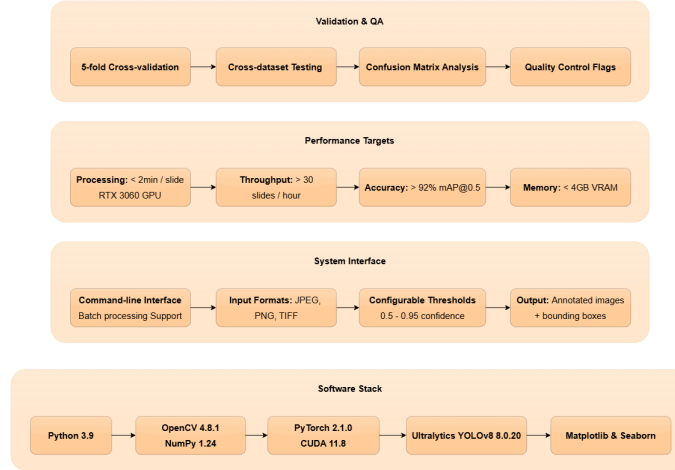


Figure 7: Implementation and Deployment Architecture.

4.8. Additional Feature: Automated Clinical Report Generation

The system can generate comprehensive clinical reports, though this component is still under development and not yet production-ready.

Proposed Report Structure:

Cell Counts: RBC, WBC subtype, and platelet counts presented in standard Complete Blood Count format, **Differential Analysis:** WBC subtype percentages and absolute counts with normal/abnormal flags, **Morphological Summary:** Aggregated statistics for cell size, shape, and cytoplasmic features, **Flagged Abnormalities:** Identified abnormalities with confidence scores, cell coordinates, and detection criteria, **Visual Elements:** Annotated source images, segmentation masks, statistical graphs, and heatmaps, **Export Formats:** PDF (archival), CSV (LIS integration), JSON (programmatic access).

Implementation Status: The reporting component requires thorough clinical validation before deployment. Current priorities focus on optimizing core detection and classification performance.

5. Dataset / Data Collection

We use three Kaggle datasets in this study to do an exhaustive analysis of blood cells. For YOLOv8 training, BCCD dataset offers 364 annotated microscopic images (640x480 resolution) with 4,888 bounding box labels covering RBC, WBC, and platelet classes in COCO format. Blood Cell Images dataset includes 12,500+ images of neutrophils, lymphocytes, monocytes, and eosinophils, to identify WBC subtypes. The ALL-IDB dataset contains 368 balanced normal/blast samples, is used for leukemia validation. Learning grows to 3,000+ images with data augmentation (rotation $\pm 15^\circ$, brightness $\pm 20\%$, flipping, elastic transforms). Each dataset is available to everyone under appropriate permissions, ensuring integrity.

Training diversity will be increased to more than 3,000 images through data augmentation (rotation $\pm 15^\circ$, brightness $\pm 20\%$, horizontal/vertical flipping, elastic deformation). Ethical compliance is ensured by the public availability of all three datasets under suitable usage licenses.

[Back to Table of Contents](#)

6. Expected Outcomes

6.1. Deliverables:

- Real-time demonstration in Python of the processing of a blood smear in less than 2 minutes using colored boxes and cell types.
- Trained YOLOv8 model with accuracy over 92% on 6-class object detection (RBC, 4WBC types, platelets) with trained
- Segmentation masks, heatmaps, and distributions for the purpose of clinical validation.
- All our code on GitHub including the trained models, image processing pipeline, and a streamlit web interface.
- Documentation with accuracy charts, error analysis, and tests on multiple datasets to prove it works
- Final reports involving confusion matrices, precision/recall curves, cross validation, and ablation experiments.

6.2. Performance Metrics:

- Classification accuracy over 92%
- Segmentation quality above 0.92 Dice score
- Throughput: Minimum 30 slides per hour

[Back to Table of Contents](#)

7. Percentage of Image Processing & Computer Vision Involvement

This project achieves **85%** Computer Vision involvement well above the 65% requirement

- **Image preprocessing:** We remove noise, normalize colors, and enhance contrast using OpenCV.
- **Cell segmentation:** Watershed algorithm and morphological operations separate overlapping cells using OpenCV and scikit-image.
- **Feature extraction:** We are measuring characteristics per cell like shape, size, color, texture by using NumPy and scikit-image.
- **Deep Learning:** YOLOv8 training and real-time inference with PyTorch identifies cell types.
- **Tools:** OpenCV 4.8.1, PyTorch 2.1.0, Ultralytics YOLOv8, NumPy, scikit-image, Streamlit.

[Back to Table of Contents](#)

References

- [1] A. Y. A. B. R. D. Oussama El Othmani, Amine Mosbah, “Ai-driven automated blood cell anomaly detection: Enhancing diagnostics and telehealth in hematology,” *National Library of Medicine National Center for Biotechnology Information*.
- [2] D. D. A. T. T. H. H. T. L. H. *Duong Trong Luong, Do Thanh Tuan and T. X. n Thang, “Detection, classification, and counting blood cells using yolov8,” *Hanoi Medical University*, 2023.
- [3] T. T. H. H. T. H. D. K. D.T. Luong, D.D. Anh, “Distinguish normal white blood cells from leukemia cells by detection, classification, and counting blood cells using yolov5,” *In IEEE 7th National Scientific Conference on Applying New Technology in Green Buildings (AtiGB)*, 2022.
- [4] X. L. F. Y. X. L. X. Z. Y. W. Xuan Chen, Linxuan Li, “Nbcde-yolov8: A new framework to improve blood cell detection and classification based on yolov8,” *The Institution of Engineering and Technology*, 2025.
- [5] R. G. J. D. T. D. J. Malik, “accurate object detectionand semantic segmentation.,” *In IEEE*, 2014.
- [6] D. E. C. S. S. R. C.-Y. F. . A. C. B. Wei Liu, Dragomir Anguelov, “Single shot multibox detector,” *Conference paper SSD*., 2016.
- [7] J. M. J. W. H. C. BORUI SUN, XIANGSUO FAN and L. LIU, “White blood cell detection based on fbdm-yolov8s,” *IEEE*, 2025.
- [8] M. K. S. S. Y. Wang, J. Zhang and X. Chen, “Self-supervised equivariant attention mechanism for weakly supervised semantic segmentation,” *IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020.
- [9] S. H. S. A. R. A. A.-B. S. d. K. Muhammad Shahzad, Farman Ali, “Blood cell image segmentation and classification: a systematic review,” *Journal of Medical Systems*, 2024.
- [10] A. Engert, “The european hematology association roadmap for european hematology research: a consensus document,,” *Haematologica*, vol. 101, no. 2, pp. 115–208, Feb. 2016, doi: 10.3324/haematol.2015.136739, 2016.